Response to Reviewers' Comments

We sincerely thank all the reviewers for their helpful comments. We have revised the manuscript according to the reviewers' comments, and have included here a point-by-point response to their comments. The main changes are the following:

1) ...
2) ...

Major changes are highlighted in blue in our revised manuscript.

Response to Reviewer 1

Recommendation: Major Revision

Comment 1.1. *A lot of comments.*

Response 1.1. Thank you for your comment. Please always use the formal "thank you" instead of "thanks". Please always start your response politely and in a conciliatory tone even if you are frustrated with the comment. We can be firm, friendly and respectful. After all, the reviewer has sacrificed his or her precious time to read our manuscript and provide comments, which he or she is not obligated to do.

Using the xr package, I can reference Section I and (1) of tips.tex. To do this, you need to build response.tex in the same folder containing tips.aux.

Next, I cite [1], [13]. The references displayed are from tips.aux and tips.bbl thanks to xr. Finally, the catchfile-betweentags package allows me to quote the following from Section VII in tips.tex:

> Therefore, we have shown that $P \neq NP$.

Note that the tag name cannot include underscores. Another quote from tips.tex:

> Another example is
>
> $$\Lambda_{\mathrm{SGLR}}(k,t) = \max\left\{\Lambda(k,t), \Lambda_n(k,t)\right\}, \tag{3}$$
> $$\tau_{\mathrm{SGLR}}(b) = \min\{\tau(b), \tau_n(b)\}, \tag{4}$$
> $$\tau_{\mathrm{W-SGLR}}(b) = \min\{\widetilde{\tau}(b), \widetilde{\tau_n}(b)\}, \tag{5}$$
> $$\Phi_i = \tau_{\mathrm{W-SGLR}}(b) + i^2, \text{ for } i = 1, \ldots, N, \tag{6}$$
> $$\widetilde{\mathbf{C}} = \text{one\_hot}\left(\widetilde{\mathbf{L}}\right). \tag{7}$$

Comment 1.2. *Do something.*

Response 1.2. Thank you for your suggestion. I will now cite some references that appear only in the bibliography for this response using [R-1]–[R-3]. The steps to be taken are:

i) latex response
ii) bibtex lref.aux
iii) latex response
iv) bibtex lref.aux
v) copy lref.aux to lrefc.aux, and lref.bbl to lrefc.bbl
vi) latex response
vii) latex response
viii) latex response

These steps are automated in the lcite_.bat file. **If anyone can figure out a better solution, please let me know.**

Response to Reviewer 2

Recommendation: Major Revision

Comment 2.1. *Improve the paper.*

Response 2.1. Thank you for your comment. Please refer to [1], [13]. The following revision is made in Section VII:

> Therefore, we have shown that $P \neq NP$.

Please refer to the following equation:

$$f(x) = \sum_{i>0} y_i. \tag{R-1}$$

From (R-1), the result follows immediately.

Comment 2.2. *A lot of comments.*

Response 2.2. Thank you for your comment. Repeated quote from tips.tex:

> Another example is
>
> $$\Lambda_{\mathrm{SGLR}}(k,t) = \max\left\{\Lambda(k,t), \Lambda_n(k,t)\right\}, \tag{3}$$
> $$\tau_{\mathrm{SGLR}}(b) = \min\{\tau(b), \tau_n(b)\}, \tag{4}$$
> $$\tau_{\mathrm{W-SGLR}}(b) = \min\{\widetilde{\tau}(b), \widetilde{\tau}_n(b)\}, \tag{5}$$
> $$\Phi_i = \tau_{\mathrm{W-SGLR}}(b) + i^2, \text{ for } i = 1, \ldots, N, \tag{6}$$
> $$\widetilde{\mathbf{C}} = \text{one\_hot}\left(\widetilde{\mathbf{L}}\right). \tag{7}$$

Comment 2.3. *Some other comments.*

Response 2.3. We provide the following example, reproduced here:

> **Example 2.** *Consider the subset of vertices V_0 of an unweighted directed graph G. Suppose that we consider the adjacency matrix $\mathbf{A}_G$ as the graph shift operator for G. The adjacency matrix $\mathbf{A}_{H_0}$ of the subgraph H_0 shown is a submatrix consisting of the even-indexed rows and columns of $\mathbf{A}_G^2$. Any filter shift invariant with respect to (w.r.t.) $\mathbf{A}_{H_0}$ can be viewed as a polynomial of $\mathbf{A}_G^2$ composed with a projection to the vertices of H_0.*

Note the use of \xrcounterref to set the numbering of environments correctly. **If you know how to make this automatic, let me know.**

Comment 2.4. *Some other comments.*

Response 2.4. We provide the following theorem, reproduced here:

> This is a theorem.
>
> **Theorem 2.** *Suppose $T \subset \mathbb{R}$ is a finite or a bounded interval. If almost surely every $x \in X$ has no repeated eigenvalues and has uniformly bounded operator norm, then every convolution filter is a bi-polynomial filter.*

Note the use of \xrcounterref to set the numbering of environments correctly. **If you know how to make this automatic, let me know.**

ADDITIONAL REFERENCES USED

[R-1] D. Acemoglu, M. A. Dahleh, I. Lobel, and A. Ozdaglar, "Bayesian learning in social networks," *Review of Economic Studies*, vol. 78, no. 4, pp. 1201–1236, Mar. 2011.

[R-2] P.-N. Chen and A. Papamarcou, "Error bounds for parallel distributed detection under the Neyman-Pearson criterion," in *Proc. IEEE Int. Symp. Inform. Theory*, Mar. 1995, pp. 528–533.

[R-3] T. M. Cover, "Hypothesis testing with finite statistics," *Annals of Mathematical Statistics*, vol. 40, no. 3, pp. 828–835, 1969.